Quadratic Simultaneous Equations

Exam code: 0607

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Updated on 9 February 2026

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Quadratic simultaneous equations

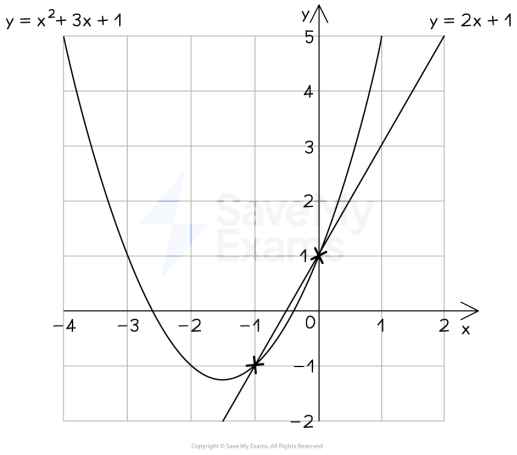
What are quadratic simultaneous equations?

- When there are two unknowns (e.g. x and y) in a problem, we need two equations to be able to find them both; these are called **simultaneous equations**
- If there is an x^2 or y^2 or xy in one of the equations then they are **quadratic** (or **non-linear**) simultaneous equations

How do I use a graph to solve quadratic simultaneous equations?

- **Plot** both equations on the same set of axes
  - To do this, you can use a table of values
  - Or for straight lines it can help to rearrange into $y = mx + c$
- Find the point where the lines **intersect**
  - The x and y **solutions** to the simultaneous equations are the x and y **coordinates** of the point of **intersection**
- E.g. To solve $y = x^2 + 3x + 1$ and $y = 2x + 1$ simultaneously

Help me understand
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



How do I solve quadratic simultaneous equations using algebra?

- Use **substitution**
  - **Substitute** the **linear** equation, *y* = ... (or *x* = ...), into the quadratic equation
    - Do not try to substitute the quadratic equation into the linear equation
- E.g. To solve $x^2 + y^2 = 25$ and $y - 2x = 5$
  - **Rearrange** the **linear** equation into $y = 2x + 5$
  - **Substitute** this into the quadratic equation, replacing all *y*'s with $(2x + 5)$
    - $x^2 + (2x + 5)^2 = 25$
- Expand and **solve** this **quadratic** equation
  - $x^2 + (2x + 5)(2x + 5) = 25$
  - $x^2 + 4x^2 + 20x + 25 = 25$
  - $5x^2 + 20x = 0$
  - $5x(x + 4) = 0$
  - $x = 0$ and $x = -4$
- Substitute **each** value of *x* into the **linear** equation, $y = 2x + 5$, to find the corresponding *y* values
  - $y = 2(0) + 5 = 5$
  - $y = 2(-4) + 5 = -3$
- Present your solutions in a way that makes it obvious which *x* belongs to which *y*
  - *x* = 0, *y* = 5 or *x* = -4, *y* = -3
- **Check** your final solutions satisfy both equations

What if the quadratic has repeated roots or no roots?

- If the resulting quadratic after substituting has a **repeated root**,
  - then the line is a **tangent** to the curve
    - i.e. the curve and the line intersect in one place only

- There are **no solutions** to the simultaneous equations
- If this happens it *may* be an indicator that your working is wrong!

What if I can't substitute one equation into the other straight away?

- If the linear equation is **not** in the form $y = \ldots$ or $x = \ldots$
  - You will need to **rearrange** it first, so that it can be **substituted** into the quadratic equation

- Consider solving $xy = 3$ and $x + y = 4$
- Either:
  - Rearrange the second equation to $y = 4 - x$ and substitute into $xy = 3$
    - $x(4 - x) = 3$
    - Expanding produces a quadratic that can be solved for x
    - $4x - x^2 = 3$
  - Or rearrange the first equation to $y = \frac{3}{x}$ and substitute into $x + y = 4$
    - $x + \frac{3}{x} = 4$
    - Multiplying both sides by x produces a quadratic that can be solved for x
    - $x^2 + 3 = 4x$

Examiner Tips and Tricks

When giving your final answer, make sure you indicate which x and y values go together.

Worked Example

Solve the equations

$$x^2 + y^2 = 36$$
$$x - 2y = 6$$

Answer:

Number the equations

$$x^2 + y^2 = 36 \quad (1)$$
$$x - 2y = 6 \quad (2)$$

There is one quadratic equation and one linear equation so this must be done by substitution

You could rearrange to make y the subject instead, but this results in a fraction which can be more tricky to deal with

Rearranging equation 2

$$x = 2y + 6$$

Substituting into equation 1

$$(2y + 6)^2 + y^2 = 36$$

Expand the brackets
Remember that a bracket squared should be treated the same as double brackets

$$(2y + 6)(2y + 6) + y^2 = 36$$
$$4y^2 + 6(2y) + 6(2y) + 6^2 + y^2 = 36$$

Simplify

$$4y^2 + 12y + 12y + 36 + y^2 = 36$$
$$5y^2 + 24y + 36 = 36$$

Rearrange to form a quadratic equation that is equal to zero
Do this by subtracting 36 from both sides

$$5y^2 + 24y = 0$$

Take out the common factor of y

$$y(5y + 24) = 0$$

Solve to find the values of y by equating each factor to zero

$$y = 0 \text{ or } 5y + 24 = 0$$

Solve the linear equation above

$$y = -\frac{24}{5}$$

So the two y values are

$$y_1 = 0$$
$$y_2 = -\frac{24}{5}$$

Substitute the values of y into one of the equations (the linear equation is easiest) to find the values of x

$$x = 2y + 6$$

$$x_1 = 2(0) + 6 = 6 \qquad x_2 = 2\left(-\frac{24}{5}\right) + 6 = -\frac{18}{5}$$

Write the final solutions in clear pairs

$$\mathbf{x_1 = 6,\ y_1 = 0}$$
$$\mathbf{x_2 = -\frac{18}{5},\ y_2 = -\frac{24}{5}}$$

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